Logistic Regression

$\hat{y} = mx + c$

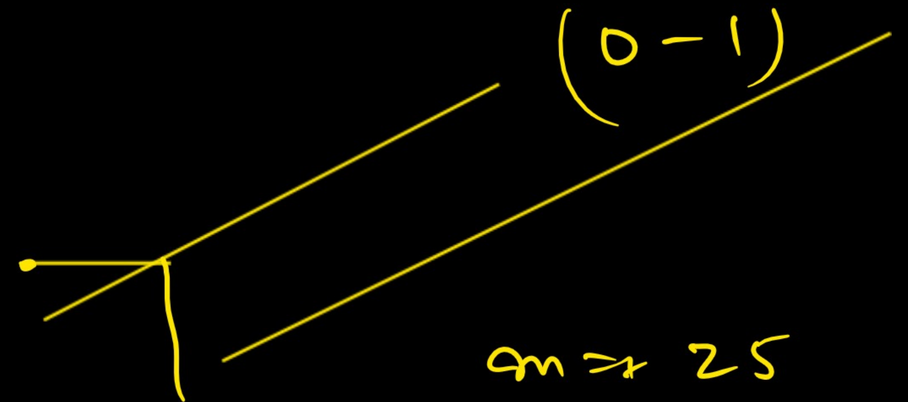
X   y

hours (X)	Score (y)	Score-P
2	50	F 0
4	60	P 1
8	80	P 1
10	95	P 1
7.5	?	

$y = mx + c$

$y = x + 1$

$y = 2 + 1 = 3$



$m \Rightarrow 25$

$c \Rightarrow 1$

$y = 25x + 1$

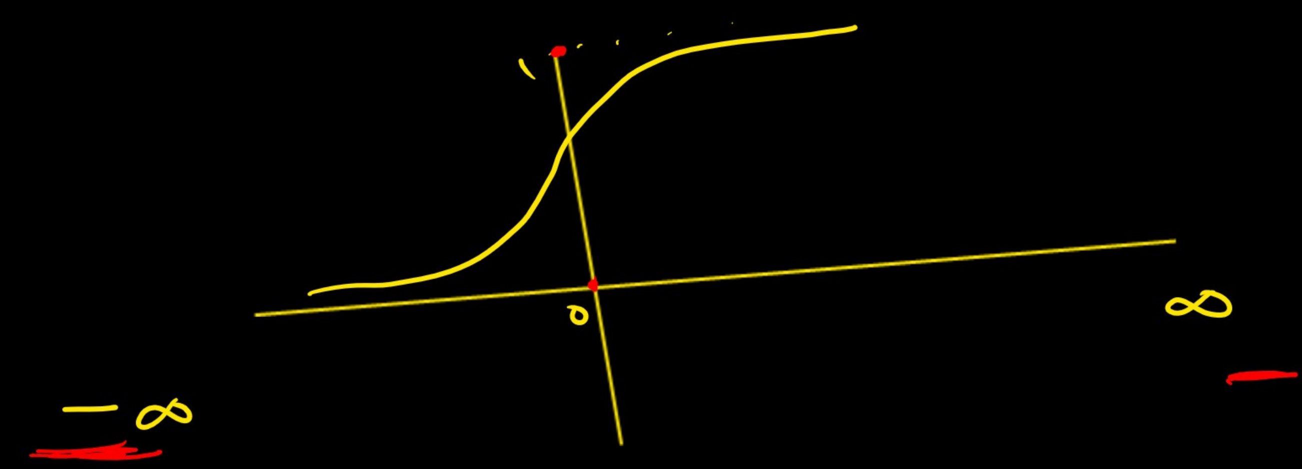
$y = 25 \times 2 + 1$

$$\boxed{y = 51}$$

$y = mx + c$ $\quad$ y $\quad$ $-\infty$ to $+\infty$ $\quad$ x

0 to 1 $\quad$ ✓

$$\sigma(y) = \frac{1}{1+e^{-y}}$$ → Sigmoid function



$$P(y=1|x) = \hat{p}$$

$$P(y=0|x) = 1-\hat{p}$$

0.8

$(1-0.8) = 0.2$

Likelihood function $L(m,c) = \hat{p}^{y}(1-\hat{p})^{1-y}$

$$P(y|p) = p^{y}(1-p)^{1-y}$$

$$= 1\,(1-1)^{1-1=0}$$

$$= 1$$

$$P(0|P) = p^{y}(1-p)^{1-y} = p \times (0)^{1-0}$$

$$= 0$$

$$L(m, c) = (p_i)^{y_i} (1-p_i)^{1-y_i}$$

$$\log(ab) = \log a + \log b$$

$$\log(x^k) = k \log x$$

$$L(m, c) = \prod_{i=1}^{n} p_i^{y_i} (1-p_i)^{1-y_i}$$

$$\log L(m, c) = \log\left(\prod \underbrace{p^y}_{a} \underbrace{(1-p)^{1-y_i}}_{b}\right)$$

$$\log L = \log p^y + \log (1-p)^{(1-y_i)}$$

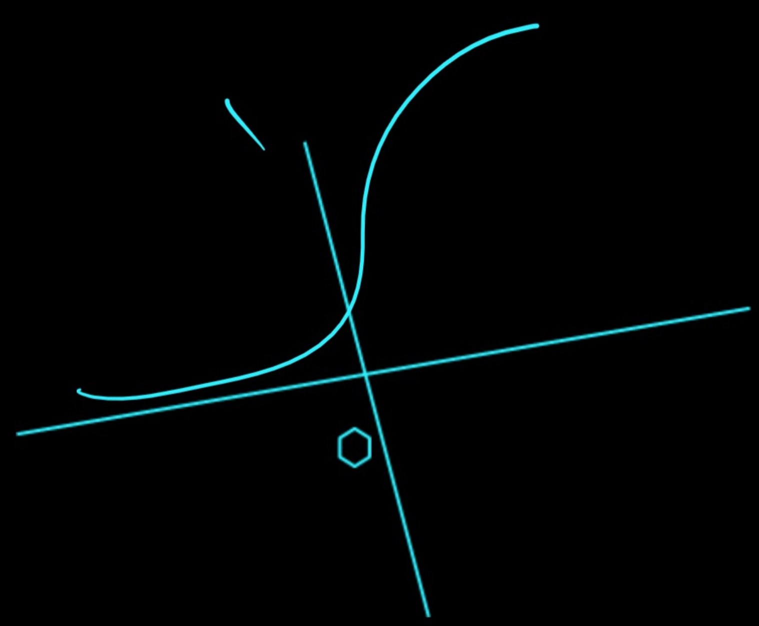
$$\log L = y \log p + (1-y_i) \log (1-p)$$

$$L(m, c) = -\frac{1}{m} \sum_{i=1}^{m} \left[y \log p + (1-y) \log (1-p) \right]$$

$$y = mx + c$$

$$m_n = m_0 - \eta \left(\frac{\partial L}{\partial m} \right) \quad \text{— (1)}$$

$$c_n = c_0 - \eta \left(\frac{\partial L}{\partial c} \right) \quad \text{— (2)}$$



$$y = mx + c$$

$$\sigma(y) = \left(\frac{1}{1 + e^{-y}} \right) = \frac{1}{1 + e^{-(mx + c)}}$$

$$\frac{\partial L}{\partial m} = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}_i - y_i)\, x$$

$$\frac{\partial L}{\partial c} = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}_i - y_i)$$